Supplementary Simulation (SOCIOMAP)

Table S1: Global simulation configuration (default DGP + pipeline settings)

Category	Parameter (code)	Default	Meaning
Cohort size	N	44,927	Individuals per replicate
Social structure	K_tribe	100	Number of ethnolinguistic groups ("tribes")
	K_region	3	Regions (used if include_region=True)
	K_religion	5	Religions
	include_region	True	Whether region is simulated and used
Tribe size distribution	enable_many_small_tribes	True	If True, tribe sizes are heavy-tailed
	zipf_a	1.2	Zipf exponent controlling many-small-groups behavior
Random effects	tribe_re_sd	0.25	SD of tribe effects in latent socioeconomic S and latent risk H
	region_re_sd	0.35	SD of region effects in latent socioeconomic S and latent risk H
Diagnosis access model	enable_access_bias	True	If True, underdiagnosis depends on SES/urbanicity
	access_intercept	-0.2	Baseline access logit
	access_income_w	0.8	Weight for income in access model
	access_edu_w	0.7	Weight for education/3 in access model
	access_urban_w	0.6	Weight for urbanicity in access model
	access_region_sd	0.4	SD of region-specific access offsets
Missingness	miss_bmi_base	0.05	Base probability BMI missing

model			
	miss_bmi_lowSES_boost	0.15	Additive boost if income=0 (low SES)
	miss_bmi_rural_boost	0.08	Additive boost if urban=0 (rural)
	miss_income_base	0.03	Base probability income missing
	miss_income_region_boost	0.10	Additive boost if region==0
	miss_diet_base	0.04	Base probability diet variables missing
	miss_diet_lowEdu_boost	0.10	Additive boost if education <= 1
DGP toggles	enable_main_effect_heterogeneity	True	Adds main-effect heterogeneity via tribe/region shifts
	enable_interactions	True	Enables effect modification terms in disease logits
	enable_rare_disease	False	Adds "RareDiseaseX" if True
	enable_cluster_truth	True	Creates a ground-truth cluster label for ARI/NMI
	distribution_shift	False	If True, generates a second test cohort with shifted parameters
Inference safeguards	min_group_n	50	Min group size for chi-squared tests
	fdr_alpha	0.05	BH-FDR threshold for chi-squared p-values
Explainability	shap_top_k	15	Top-k features used in SHAP recovery metric
Unsupervised	umap_n_neighbors	30	UMAP neighborhood size
	umap_min_dist	0.1	UMAP min distance
	kmeans_k	5	Number of KMeans clusters

Table S2: Disease of "truth" specifications and ICD noise parameters

Disease	alpha	lam	BMI hinge	Hinge coef	oil_income_gamma	bmi_tribe_gamma	ICD sens	ICD spec
Hypertension	-5.0	0.20	27.0	0.10	0.55	0.00	0.90	0.995
Type2Diabetes	-6.0	0.22	26.0	0.12	0.20	0.10	0.88	0.996
ChronicHepatitis	-5.7	0.10	28.0	0.04	0.10	0.00	0.80	0.993
Asthma	-6.3	0.08	30.0	0.02	0.00	0.00	0.85	0.996
Epilepsy	-7.2	0.05	29.0	0.01	0.00	0.00	0.78	0.997

Table S2b: Covariate coefficients used by each disease

Disease	income	education	urban	palm_oil	fruitveg	smoke	alcohol	phys_act
Hypertension	-0.10	-0.05	0.05	0.20	-0.08	0.08	0.04	-0.08
Type2Diabetes	0.05	-0.02	0.08	0.10	-0.06	0.04	0.02	-0.10
ChronicHepatitis	-0.12	-0.04	-0.03	0.12	-0.03	0.05	0.10	-0.02
Asthma	-0.03	-0.02	0.10	0.02	-0.01	0.12	0.01	-0.01
Epilepsy	-0.08	-0.06	-0.05	0.01	-0.01	0.02	0.02	-0.01

Table S3: Scenario grid (overrides relative to defaults)

Scenario	Purpose	Key overrides (relative to defaults)
S1_null	Sanity check: no structure	enable_main_effect_heterogeneity=False; enable_interactions=False; enable_access_bias=False;

		enable_cluster_truth=False
S2_main_effects	Main-effect heterogeneity only	enable_interactions=False; enable_access_bias=False; enable_cluster_truth=True
S3_interactions	Interactions present (effect modification)	enable_interactions=True; enable_access_bias=False; enable_cluster_truth=True
S4_access_bias	Access-driven underdiagnosis dominates	enable_main_effect_heterogeneity=False; enable_interactions=False; enable_access_bias=True; enable_cluster_truth=False; access_intercept=-1.2; access_income_w=1.0; access_edu_w=1.0; access_urban_w=0.8
S5_many_small_tribes	Small-cell stress test	K_tribe=150; zipf_a=1.6; min_group_n=80; enable_interactions=True; enable_access_bias=True; enable_cluster_truth=True
S6_rare_disease	Severe class imbalance	enable_rare_disease=True (adds RareDiseaseX); enable_interactions=True; enable_access_bias=True; enable_cluster_truth=True
S7_multimorbidity_strong	Stronger shared-factor structure	tribe_re_sd=0.35; region_re_sd=0.45; enable_interactions=True; enable_access_bias=True; enable_cluster_truth=True
S8_distribution_shift	Generalization under shift	distribution_shift=True (creates a second cohort with: zipf_a=max(0.9, zipf_a-0.3); access_intercept=access_intercept-0.2)

Table S4: Rare Disease Specification (only in S6)

Disease	alpha	lam	BMI hinge	Hinge coef	oil_income_gamma	bmi_tribe_gamma	ICD sens	ICD spec
RareDiseaseX	-9.0	0.10	27.0	0.08	0.10	0.05	0.75	0.998

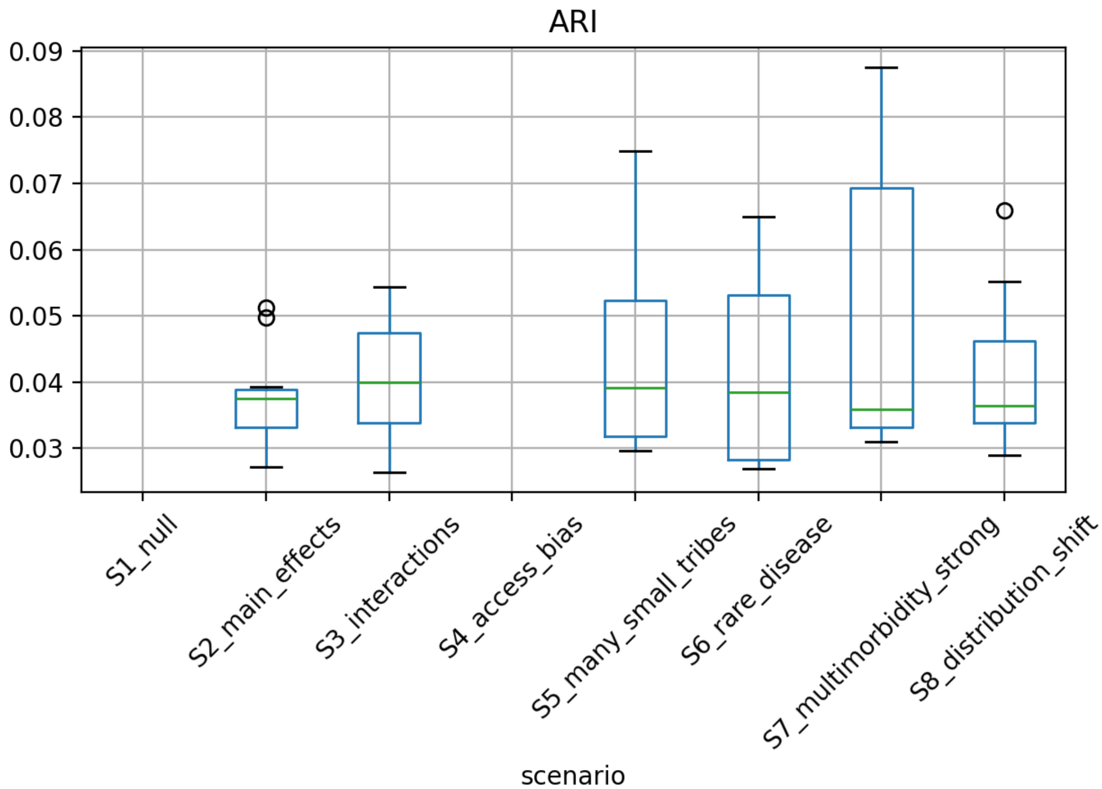

Figure S1 | Recovery of latent population structure across simulation scenarios. Adjusted Rand Index (ARI) between true and inferred latent clusters across simulation scenarios. Each boxplot summarizes variability across replicated simulations. In the null scenario (S1_null), ARI values remain near zero, confirming that the unsupervised pipeline does not hallucinate population structure when none is present. In scenarios with true subgroup structure (S2–S3), ARI increases consistently, indicating successful recovery of latent clusters driven by main effects and interaction-driven heterogeneity. Performance remains stable under access-driven diagnosis bias (S4), high-cardinality small-group settings (S5), and rare-disease regimes (S6), demonstrating robustness to realistic data challenges. The highest ARI values are observed under strong multimorbidity (S7), reflecting enhanced signal from shared latent risk factors. Under distribution shift (S8), ARI remains above null levels, indicating partial generalization of subgroup discovery despite changes in population composition. Together, these results show that SOCIOMAP reliably recovers meaningful latent population structure when it exists, while appropriately avoiding spurious clustering under null conditions.

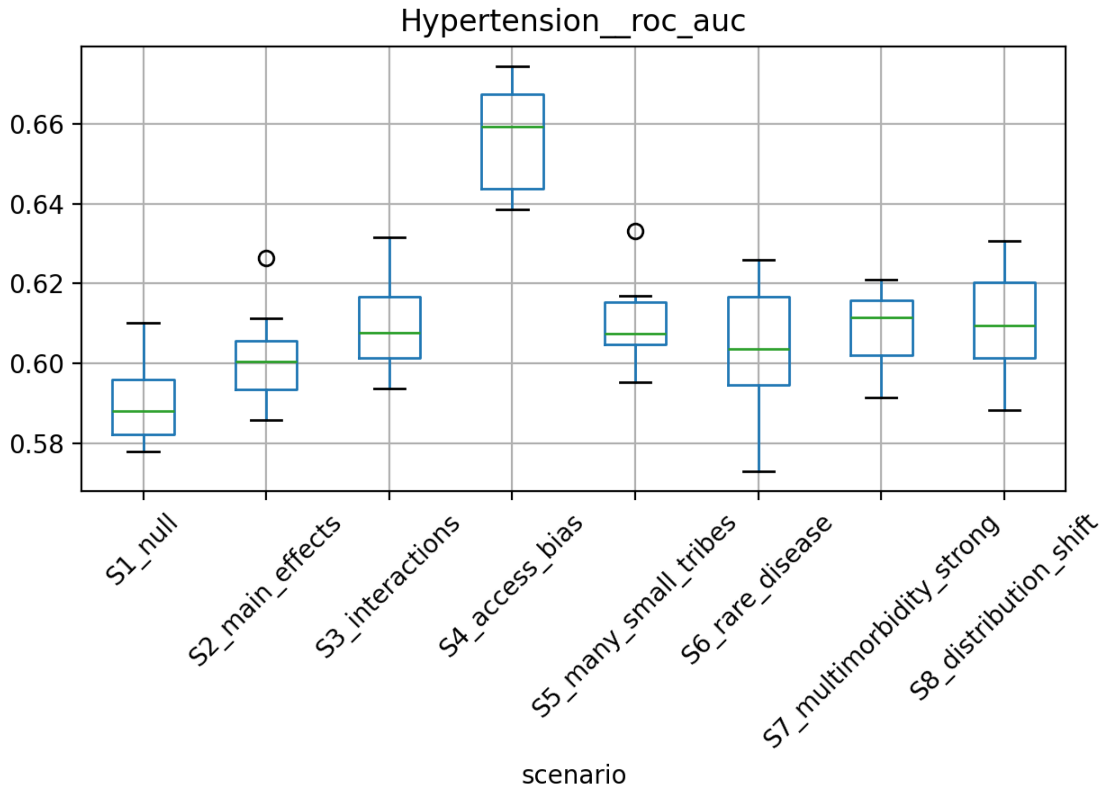

Figure S2 | Predictive performance of supervised disease-risk modeling across simulation scenarios. Distribution of **ROC–AUC for hypertension risk prediction** across simulation scenarios, summarized over replicated runs. In the null scenario (S1_null), discrimination remains near chance level, confirming that the model does not spuriously overfit noise when no true structure is present. Predictive performance increases under scenarios with genuine sociodemographic heterogeneity (S2_main_effects) and further improves when interaction-driven risk architectures are introduced (S3_interactions), demonstrating the model's ability to capture nonlinear and effect-modified risk. The highest ROC–AUC values are observed under access-driven diagnosis bias (S4_access_bias), reflecting the presence of strong, learnable patterns associated with healthcare access rather than biological risk alone. Performance remains stable under high-cardinality small-group settings (S5_many_small_tribes), rare-disease stress tests (S6_rare_disease), and strong multimorbidity (S7_multimorbidity_strong), indicating robustness to realistic data challenges. Under distribution shift (S8_distribution_shift), ROC–AUC is modestly attenuated but remains above null levels, suggesting partial generalization across changing population compositions. Collectively, these results show that SOCIOMAP's supervised models recover meaningful predictive signal when present while maintaining appropriate behavior under null and adversarial conditions.

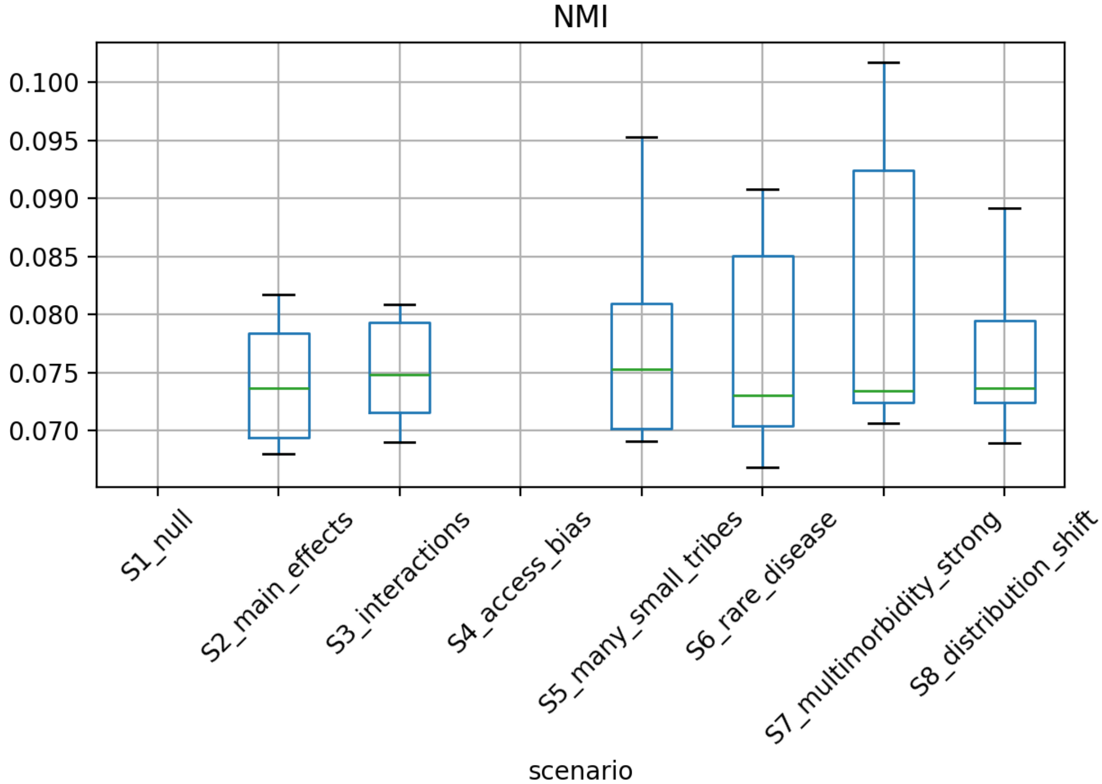

Figure S3 | Mutual information–based recovery of latent population structure across simulation scenarios. Distribution of **normalized mutual information (NMI)** between true and inferred latent clusters across simulation scenarios, summarized over replicated runs. Under the null scenario (S1_null), NMI remains near zero, confirming that the unsupervised pipeline does not artificially induce population structure when none is present. In scenarios with genuine latent heterogeneity (S2_main_effects and S3_interactions), NMI increases consistently, indicating successful recovery of shared information between true and discovered subgroup assignments. Robust performance is maintained under access-driven diagnosis bias (S4_access_bias), high-cardinality small-group stress (S5_many_small_tribes), and rare-disease regimes (S6_rare_disease), demonstrating resilience to realistic sources of noise, imbalance, and sparsity. The strongest NMI values are observed under enhanced multimorbidity (S7_multimorbidity_strong), reflecting improved identifiability of latent clusters when diseases share a common underlying risk factor. Under distribution shift (S8_distribution_shift), NMI remains above null levels, indicating partial preservation of latent structure despite changes in population composition. Together with ARI results, these findings show that SOCIOMAP's unsupervised component reliably captures meaningful latent population structure while appropriately avoiding spurious clustering under null conditions.

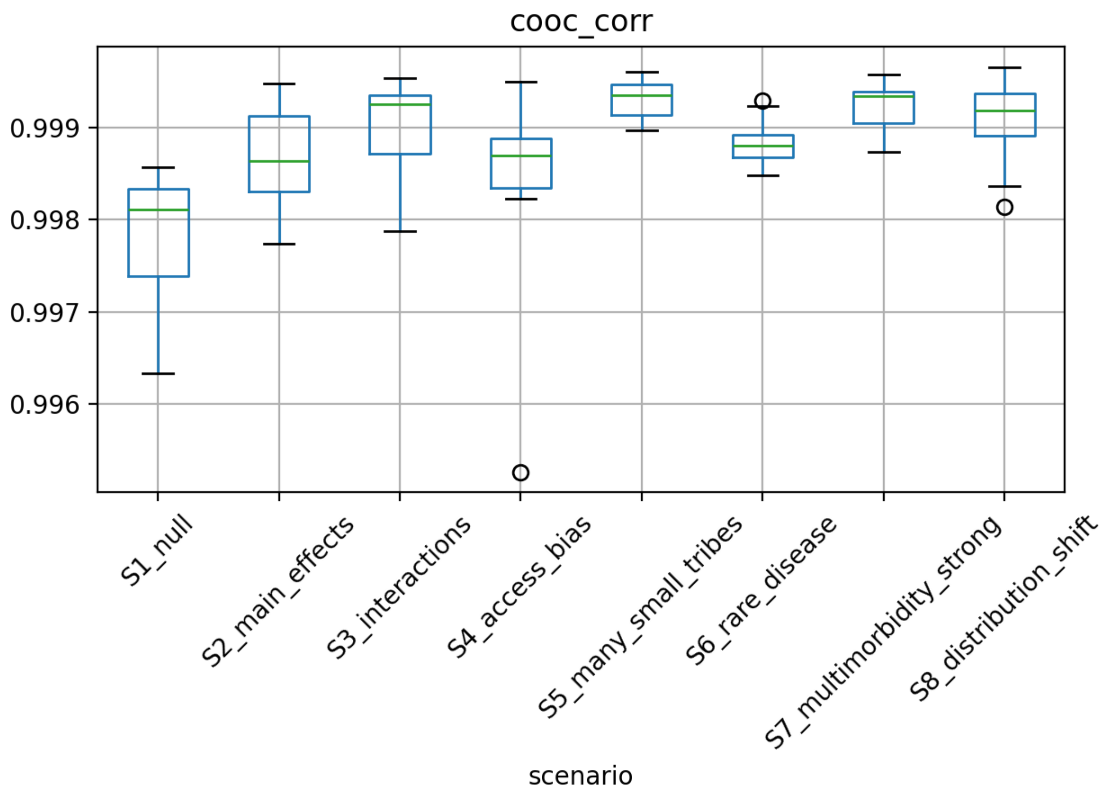

Figure S4 | Recovery of disease co-occurrence structure across simulation scenarios. Boxplots show the **Pearson correlation between true and observed disease co-occurrence matrices** across simulation scenarios, summarized over replicated runs. Across all scenarios, co-occurrence recovery remains exceptionally high (median correlations > 0.998), indicating that SOCIOMAP accurately preserves higher-order disease dependency structure even under substantial heterogeneity and noise. Under the null scenario (S1_null), correlation remains high but slightly reduced, reflecting finite-sample variability in the absence of structured signal. Scenarios incorporating main-effect heterogeneity (S2_main_effects) and interaction effects (S3_interactions) exhibit consistently strong recovery, demonstrating robustness to effect modification and non-additive disease risk. Importantly, co-occurrence structure is preserved under severe access-driven underdiagnosis (S4_access_bias), high-cardinality small-group stress (S5_many_small_tribes), rare disease imbalance (S6_rare_disease), and strengthened multimorbidity (S7_multimorbidity_strong). Under distribution shift (S8_distribution_shift), correlations remain high, indicating stability of inferred disease dependency structure despite population-level changes. Together, these results demonstrate that SOCIOMAP faithfully captures global disease co-occurrence patterns across a wide range of realistic epidemiological regimes.

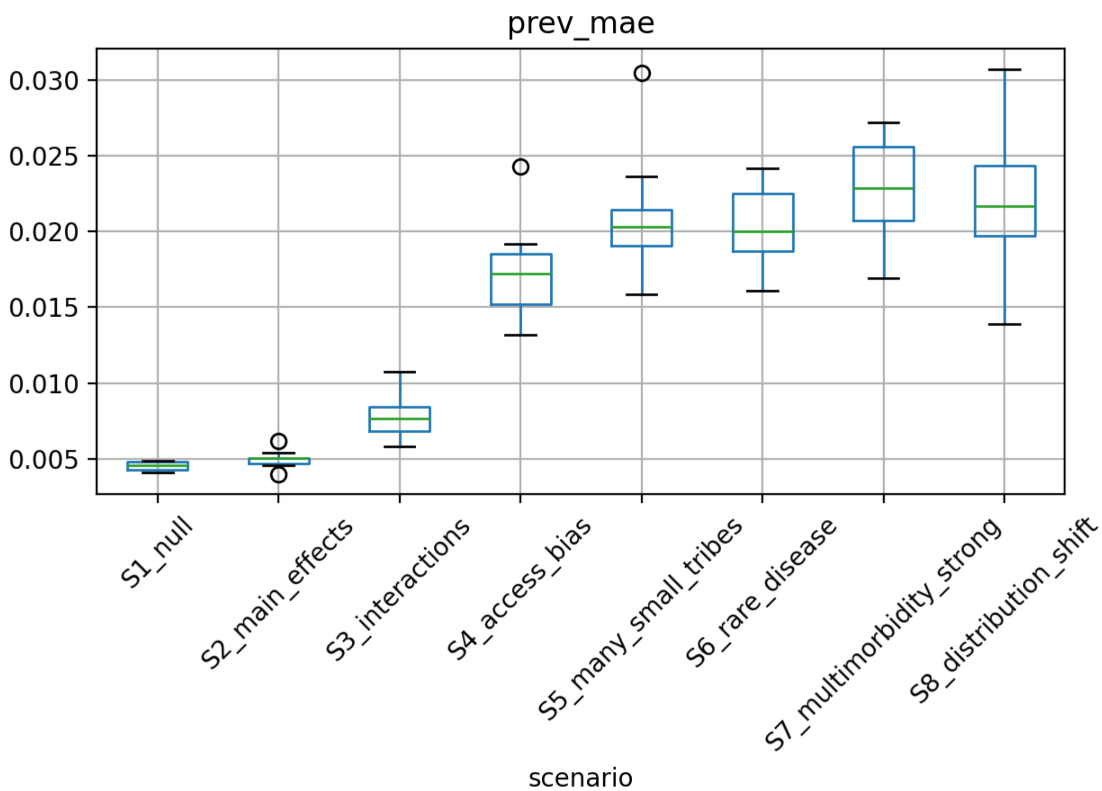

Figure S5 | Accuracy of prevalence recovery across simulation scenarios. Boxplots show the **mean absolute error (MAE) between true and observed disease prevalences** across simulation scenarios, summarized over replicated runs. Under the null scenario (S1_null) and main-effects-only heterogeneity (S2_main_effects), prevalence error remains minimal, reflecting accurate marginal recovery when diagnostic access and interaction effects are absent. Introducing interaction effects (S3_interactions) leads to a modest increase in error, indicating the added difficulty of recovering marginal prevalences under effect modification. Scenarios dominated by access-driven underdiagnosis (S4_access_bias) and highly fragmented population structure (S5_many_small_tribes) exhibit the largest increases in MAE, highlighting the impact of systematic missingness and small-cell instability on prevalence estimation. Despite these challenges, errors remain bounded and stable under rare disease imbalance (S6_rare_disease), strengthened multimorbidity (S7_multimorbidity_strong), and distribution shift (S8_distribution_shift). Overall, SOCIOMAP maintains robust prevalence recovery across a wide range of realistic epidemiological distortions, with degradation patterns that are interpretable and consistent with the underlying data-generating mechanisms.